

Recovering and Redirecting Peptide Backbone Conformation via Contact-Conditioned $C\alpha$ Generation in a Cross-Reactive pMHC/TCR System

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Motivation. Some T-cell receptors are cross-reactive: they recognize several different peptides by letting each peptide fold into a different shape inside the same MHC groove [2]. This cross-reactivity is a central safety problem for TCR-based therapies, where an engineered receptor can cause dangerous off-target effects by recognizing an unintended self-peptide that adopts a compatible backbone shape. Deep generative models can now build protein backbones with high accuracy (RFdiffusion [1]), which raises a question that bears directly on that problem: given only the receptor side of the interaction, can such a model be steered to build a peptide backbone in one chosen target shape, even a shape that is not the one closest to its input? We test this in a well-studied two-peptide system [2]: the DMF5 receptor on HLA-A*02:01 binds two 10-residue peptides, GIG (6AM5) and DRG (6AMU), whose backbones differ by 2.87 Å almost entirely at the tail end (the first four positions match within 1.14 Å; position 6 moves 5.62 Å). Which register a peptide takes is set by which residue drops into the MHC F-pocket (position 10 for GIG, position 9 for DRG).

Methods. Conditioning only on receptor-side contact residues, with no peptide-backbone template, we asked whether RFdiffusion [1] can generate a $C\alpha$ backbone that recovers a peptide’s own native register or crosses into the alternate one. We generated 300 designs per crystal for 20 different conditioning schemes, plus a focused N-terminus-seeding experiment. Designs are scored against a three-criterion register test calibrated not to a single crystal but to **native molecular-dynamics ensembles** (Amber ff19SB, TIP3P, physiological **310 K**, 50 ns): $C\alpha$ proximity within a moving-block-bootstrap acceptance band (DRG within 1.48 Å, 95% CI 1.36 to 1.58), correct F-pocket occupant identity, and native burial depth. We additionally correct for a forward/reverse backbone-threading artifact that raw RMSD cannot detect.

Results. Both recovery and redirection are achievable but rare, and are gated by the breadth of conditioning rather than by any specific curated residue set. Across the de novo corpus, 4 of 8,722 designs recover the native DRG register (best 1.07 Å) and 3 of 8,722 cross into the alternate DRG register (best 1.44 Å), all forward-threaded, seating the correct p9 anchor, and from the broadest schemes. Critically, our no-conditioning control produced **zero** designs that dock in the groove or come close in RMSD to either native backbone: **0 of 932** unconditioned draws entered the band or seated an anchor, so these events are real signal above a zero-

information background, not noise. A templating control that supplies backbone geometry directly recovers register on demand (best 0.10 Å), whereas contact conditioning alone only weakly determines it, locating register in a confined tail-end signal. These results, from a single well-characterized cross-reactive system, map the current capability boundary of contact-conditioned backbone generation, separate what it can retrieve from what must still be supplied as explicit geometry, and mark a step toward anticipating the cross-reactive, off-target peptide recognition that constrains the safety of TCR-based therapies.

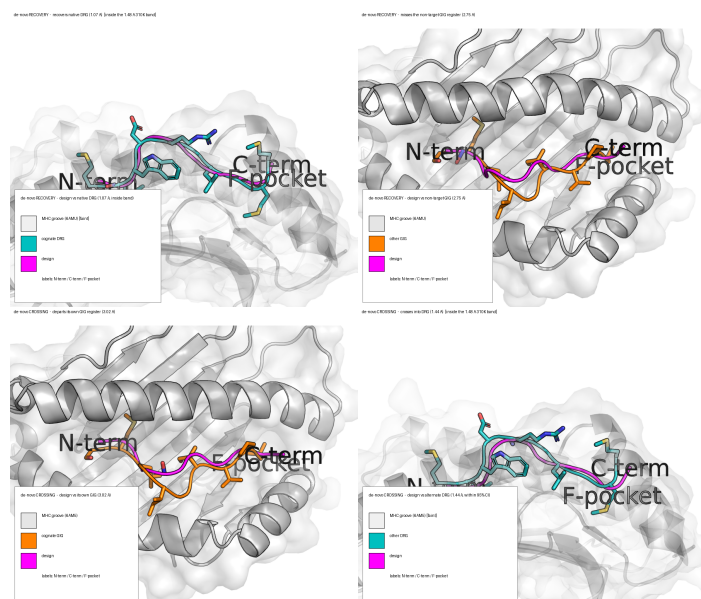


Table 1: De novo and control designs, scored against the 310 K native envelope. Recoveries land in the peptide’s own native register; crossings land in the alternate register.

design group	<i>n</i>	best RMSD own (Å)	recov.	cross.
de novo (receptor contacts)	8,722	1.07	4	3
null (no conditioning)	932	2.05	0	0
templated (backbone given)	3,060	0.10	register on demand	
crossover (N-term given)	338	0.54	16	20

References. [1] Watson JL et al. De novo design of protein structure and function with RFdiffusion. *Nature* 620:1089–1100 (2023). [2] Riley TP et al. T cell receptor cross-reactivity expanded by dramatic peptide-MHC adaptability. *Nat. Chem. Biol.* 14:934–942 (2018).

Availability. Code, the MD-calibrated scorer, campaign drivers, and curated design archives: github.com/sermare/if-mhc.